Febrile convulsions

Saturday, January 8, 2022 2:26 PM

Overview
- Convulsion due to rapid rise in temperature due to extra cranial infections
- The most common seizure disorder during childhood
- Recurrent in 30-50% of cases
- occur in 2%-5% of children
- Family history in about 20 % of cases

Pathophysiology
- Fever increase the metabolic rate of cerebral neurons → lowering the seizure threshold
- the developing brain cannot withstand rapid and large increases in temperature

Clinical features

VS	Typical simple FC	Atypical complex FC
Age	6 month - 5 yrs	< 6 month > 5 yrs
Developmental Hx	Normal	Delayed milestone
FH of epilepsy	-ve	+ve
temperature	39°C or more	>38.5°C
Duration	< 15 minute	> 15 minute
Frequency	One fit	Recurrent
character	G. tonic-clonic	Focal

Diagnostic criteria
- Age: from 6 months & below5 years
- Temperature: > 39°C
- fits occur within 8-12 hr from the onset of fever
- No evidence of CNS infection
  - Meningitis
  - abscess
- Evidence of extra cranial infection
  - Tonsilitis
  - otitis media

Differential diagnosis
- Intracranial infections e.g Meningitis, brain abscess
- Epileptic fit precipitated by associated fever
- Dravet syndrome

risk factors of recurrence
- Age < 1 yr
- Duration of fever < 24 hr
- Fever 38-39 C°

Risk of subsequent epilepsy
- Neurodevelopmental abnormalities
- Complex febrile seizures (focal)
- Family history of epilepsy

Investigations
- Lumbar puncture to rule out meningitis
- EEG & Neuro imaging for cases with ↑ risk of epilepsy
- Blood tests e.g. electrolytes, blood glucose

Management

Admission
- first fit
- Complex seizure
- child unwell

ABC
- Patent airway

seizure control
- Seizures are typically self-limited
- give rectal diazepam if the fit lasts > 5 min
- then follow seizure protocol

Control the fever
- Tepid sponging or cold bath
- paracetamol or ibuprofen

Treatment of underlying cause
- suspected infection give antibiotics
- herpetic lesions or contact, give acyclovir

Give parental advice
- Acute handling of seizures at home

Notes
- The primary reason for distinguishing simple from complex seizures is to predict the future risk of epilepsy
- lorazepam, or midazolam are alternative to diazepam